FROZEN

exam-tracker-svc

EduPath :: Exam Tracker Service

14/14 Tests Passing · JEE · NEET · CUET · Syllabus Tracking

PLATFORM	STACK	ARCHITECTURE	GENERATED
EduPath v1.0	Java 17 · Spring Boot 3.3.5	Hexagonal · Kafka · PostgreSQL	March 07, 2026

BUILD STATUS: SUCCESS · ALL TESTS PASSING · RECORD FROZEN

exam-tracker-svc — COMPLETION RECORD

STATUS: FROZEN — DO NOT MODIFY
Tests passed: 14/14 | Build: SUCCESS | Date: 2026-03-07

Service Overview

EduPath Student Career Portal — competitive exam lifecycle management service. Tracks student enrollments in JEE/NEET/CUET/UPSC exams, manages syllabus module coverage, records mock test attempts with score analytics, and logs study sessions. Publishes Kafka events for performance-svc and ai-mentor-svc consumption.

Test Results

```
Tests run: 14, Failures: 0, Errors: 0, Skipped: 0

BUILD SUCCESS — Total time: 10.976 s
```

TEST CLASS	TESTS	RESULT
ArchitectureRulesTest	5	PASS
ExamEnrollmentServiceTest	5	PASS
MockTestServiceTest	4	PASS

Architecture

Standard hexagonal (ports & adapters). `domain` has zero Spring dependencies.

```
com.edutech.examtracker/
  ExamTrackerApplication.java
  api/
    ExamEnrollmentController.java      REST — /api/v1/exam-tracker/enrollments/**
    SyllabusController.java            REST — /api/v1/exam-tracker/syllabus/**
    MockTestController.java            REST — /api/v1/exam-tracker/mock-tests/**
    StudySessionController.java        REST — /api/v1/exam-tracker/study-sessions/**
    GlobalExceptionHandler.java        RFC 7807 ProblemDetail
  application/
    service/
      ExamEnrollmentService.java       EnrollInExamUseCase, GetEnrollmentUseCase
      SyllabusService.java             GetSyllabusProgressUseCase, UpdateSyllabusModuleUseCase
      MockTestService.java             RecordMockTestUseCase, GetMockTestHistoryUseCase
      StudySessionService.java         RecordStudySessionUseCase, GetStudySessionsUseCase
    dto/                               Java records (all immutable)
    exception/                         DuplicateEnrollmentException, EnrollmentNotFoundException, M
oduleNotFoundException
  domain/
    model/
      ExamEnrollment.java              @Entity, static factory, @Version, soft delete
      SyllabusModule.java              @Entity, ModuleStatus enum (NOT_STARTED, IN_PROGRESS, COMPLE
TED)
      MockTestAttempt.java             @Entity, records score/percentile/rank
      StudySession.java                @Entity, SessionType (READING, PRACTICE, REVISION, MOCK)
    model/enums/                       ExamCode (JEE_MAIN, JEE_ADV, NEET, CUET, UPSC_CSE, CAT), Exa
mStatus, ModuleStatus, SessionType
    event/                             ExamEnrolledEvent, MockTestCompletedEvent, StudySessionRecor
dedEvent, SyllabusModuleUpdatedEvent
    port/in/                           8 use-case interfaces
    port/out/                          ExamEnrollmentRepository, SyllabusModuleRepository, MockTest
AttemptRepository, StudySessionRepository, ExamTrackerEventPublisher
  infrastructure/
    persistence/                       SpringData*Repository (package-private) + public *Persistenc
eAdapter @Component
    kafka/ExamTrackerKafkaAdapter.java  KafkaTemplate publisher, best-effort
```

```
    security/SecurityConfig.java        Trusts X-User-Id / X-User-Role headers (no JWT re-validatio
n)
    config/KafkaTopicProperties.java    @ConfigurationProperties("kafka.topics")
```

Flyway Migrations

VERSION	SCRIPT	DESCRIPTION
V1	V1__create_examtracker_schema.sql	CREATE SCHEMA examtracker_schema
V2	V2__create_exam_enrollments.sql	exam_enrollments table + unique (student_id, exam_code) partial index
V3	V3__create_syllabus_modules.sql	syllabus_modules table
V4	V4__create_mock_test_attempts.sql	mock_test_attempts table
V5	V5__create_study_sessions.sql	study_sessions table

API Endpoints

METHOD	PATH	USE CASE
POST	`/api/v1/exam-tracker/enrollments`	Enroll student in exam
GET	`/api/v1/exam-tracker/enrollments/{studentId}`	Get student enrollments
GET	`/api/v1/exam-tracker/enrollments/{id}`	Get enrollment by ID
GET	`/api/v1/exam-tracker/syllabus/{enrollmentId}`	Get syllabus progress
PATCH	`/api/v1/exam-tracker/syllabus/{moduleId}`	Update module status
POST	`/api/v1/exam-tracker/mock-tests`	Record mock test attempt
GET	`/api/v1/exam-tracker/mock-tests/{studentId}/{enrollmentId}`	Get mock test history
POST	`/api/v1/exam-tracker/study-sessions`	Record study session
GET	`/api/v1/exam-tracker/study-sessions/{studentId}`	Get study sessions

Kafka Events Published

EVENT	TOPIC	TRIGGER
ExamEnrolledEvent	`exam.tracker.enrolled`	Exam enrollment
MockTestCompletedEvent	`exam.tracker.mock.completed`	Mock test recorded
StudySessionRecordedEvent	`exam.tracker.study.session`	Study session logged
SyllabusModuleUpdatedEvent	`exam.tracker.syllabus.updated`	Module status changed

Key Design Decisions

- **Duplicate enrollment guard**: `DuplicateEnrollmentException` on `(student_id, exam_code)` collision (partial unique index with `deleted_at IS NULL`).
- **Syllabus coverage**: Computed as `completedModules / totalModules * 100` for `GetSyllabusProgressUseCase`.
- **Mock test trend**: `MockTestService` computes rolling average over last N attempts for trend score — fed to performance-svc via Kafka.
- **Security**: Trusts `X-User-Id` / `X-User-Role` headers from student-gateway. No JWT re-validation.
- **No Lombok**: Manual constructors. Constructor injection only.
- **Optimistic locking**: `@Version` on all mutable entities.

ArchUnit Rules (5)

1. `domain` package has no dependencies on Spring Framework
2. `application` services depend on `domain` ports only (no infrastructure)
3. `@Entity` classes reside in `domain.model` package
4. `@Service` classes reside in `application.service` package
5. `infrastructure` adapters implement domain port interfaces

This document is permanently frozen. Any future changes to exam-tracker-svc require a new versioned record.